



جامعة الكفيل
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Oral Surgery

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Exodontia

Third stage

Lec. 13



Exodonia (extraction)

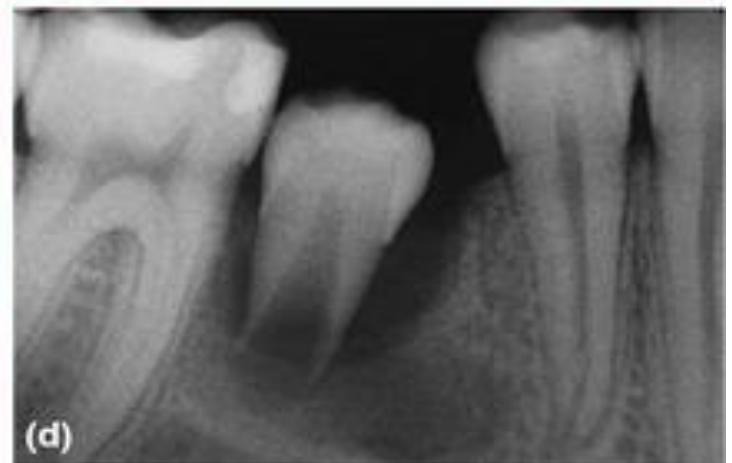
- is a **painless** removal of teeth from their bony alveolar socket with **relatively minimal amount of trauma** to the investing or surrounding tissues, so that the **wound heals** **without postoperative complications.**

Basically, we have two methods of extraction

- 1- Intra-alveolar extraction (forceps or simple extraction): consist of removing the tooth or root by the use of forceps or elevators or both, which is enough for extraction in most of cases.



- 2- Trans-alveolar extraction (surgical extraction): in trans-alveolar extraction we dissect the tooth or root from its bony attachment by rising a flap and by removal some of the bone surrounding the roots, which are then removed by the use of elevators and/or forceps.



INDICATIONS OF EXTRACTION

- There are many indications for tooth extraction, if **conservative treatment has either failed or is not indicated**, a tooth may be extracted. The reasons for extraction of teeth are based on, the presence of local pathology, the feasibility of restorative procedure, function of the dentition as a whole and the patient's attitude and education.

INDICATIONS OF EXTRACTION

- Although the modern dentistry tries and takes all measures to preserve and maintain teeth in the oral cavity, there are still a variety of general indications for removing teeth.

INDICATIONS OF EXTRACTION

- **1- Sever caries**
- This is the **most common** reason for a tooth removal. Badly carious teeth resulted in bad oral hygiene and bad smell , also the sharp edges of the carious teeth may lead to repeated ulceration to mucosa, in addition to spontaneous pain or pain during eating and drinking.
- Untreated teeth with caries may **end with pulpitis, periapical pathology etc..**



- **2- Sever periodontal disease**
- Sever and extensive periodontal disease is a **common reason** for tooth removal, in chronic periodontitis there is **excessive bone loss and mobility of the tooth.**

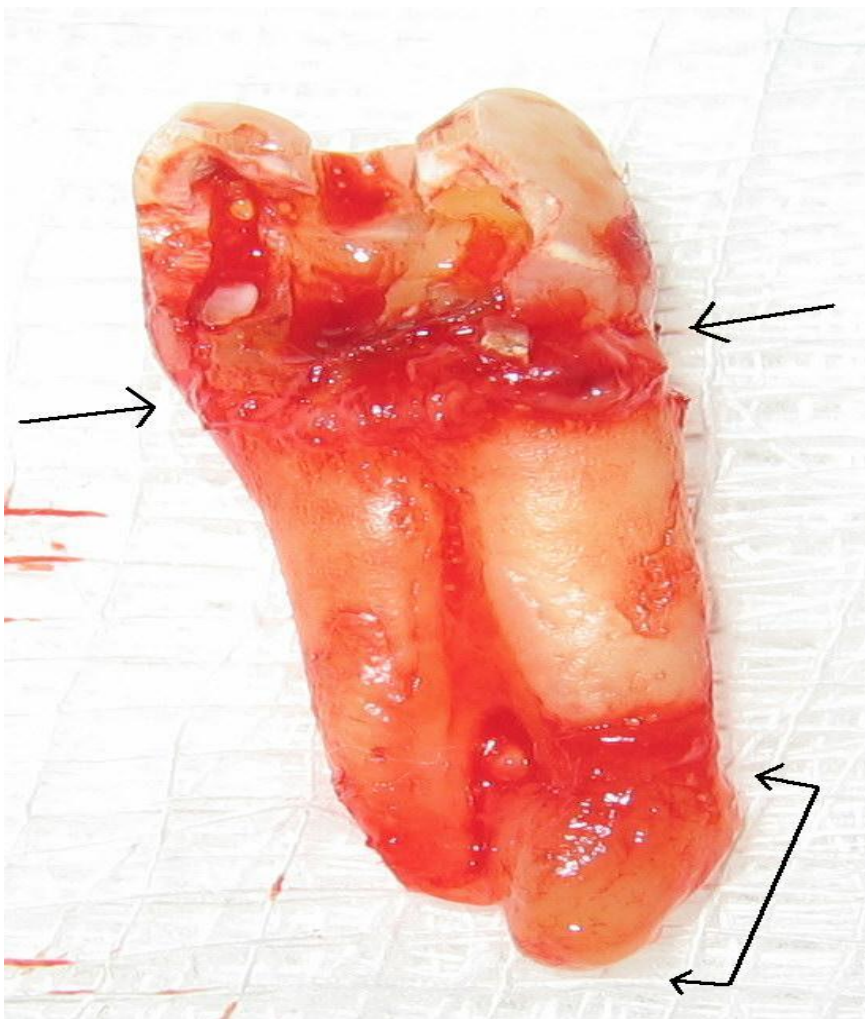
- As a rough guide, **loss of about half of the normal alveolar bone or extension of pockets to the bifurcation** of the roots of posterior teeth and hypermobility of the teeth means that extraction of the involved tooth is necessary.
- Patient with advanced periodontitis may complain of mild to severe throbbing pain in case of development of Paradental abscess.



- **3- Pulp pathology**

- For example acute or chronic pulpitis and non treatable pulpal lesion. If endodontic therapy wasn't possible or if the tooth isn't amenable for endodontic treatment, so extraction is indicated.

- **4-Apical pathology**
- Periapical lesion like periapical abscess, granuloma and cyst.
- If the teeth failed to respond to all conservative treatment to resolve periapical pathology so they are indicated for extraction.





- **5-Orthodontic reasons**
- During the course of orthodontic treatment, tooth or teeth maybe extracted for:-
- a) **Therapeutic extraction** e.g. extraction of upper **4** for treatment of malposed upper **3**.
Extraction of teeth to provide **space** for teeth alignment.

- b) **Malposed teeth** : teeth which **erupted out of arch line**, when they are difficult to clean and not amenable for orthodontic treatment.
- c) **Preventive exaction**: means that **during mixed dentition**(permanent and deciduous teeth) period, dentist may extract few deciduous teeth **to prevent future malocclusion**.



- **6- Prosthetic considerations**

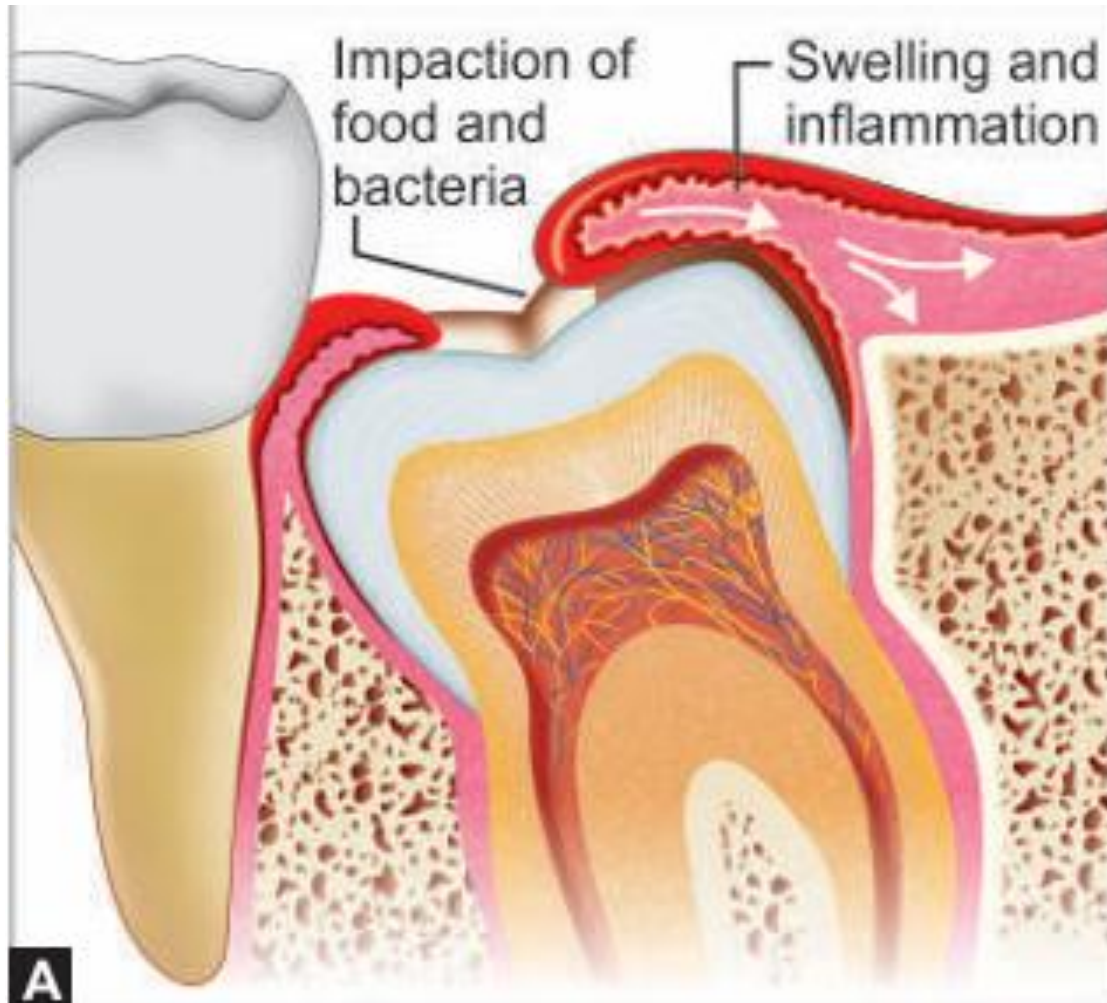
- Extraction of teeth are indicated :

- 1) Providing efficient dental prosthesis.
- 2) To provide better design for success of partial denture, few teeth maybe extracted.
- 3) Solitary tooth or non-strategic tooth to enable the patient to have complete denture e.g. full mouth clearance.

- **7- Impacted teeth**
- Retention of unerupted tooth beyond the normal time of eruption. The impacted teeth are indicated for extraction when they responsible for:
 - a) Vagus facial pain.
 - b) Periodontal problems of the adjoining teeth.

- c) Temporo-mandibular joint problems
- d) Bony pathology eg cyst (dentigerous cyst), tumor.
- e) May predispose, to anterior teeth crowding.

f) Significant infection (pericoronitis) e.g. partially erupted third molar.

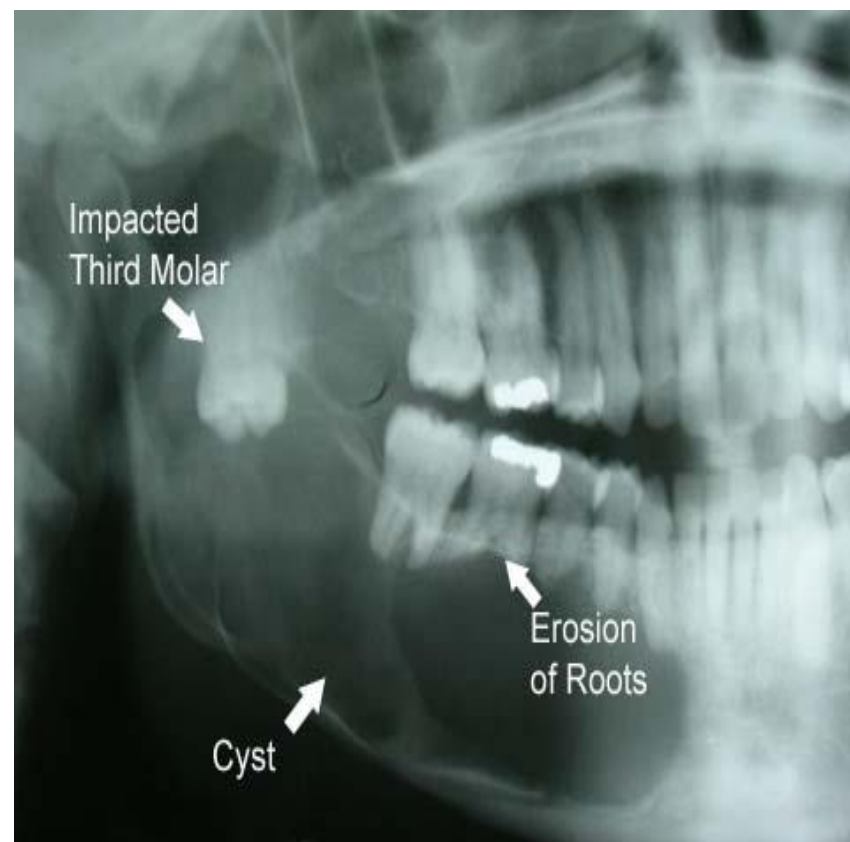


Note the food debris collecting under the gum tissue.
The slight redness is the beginning of pericoronitis.

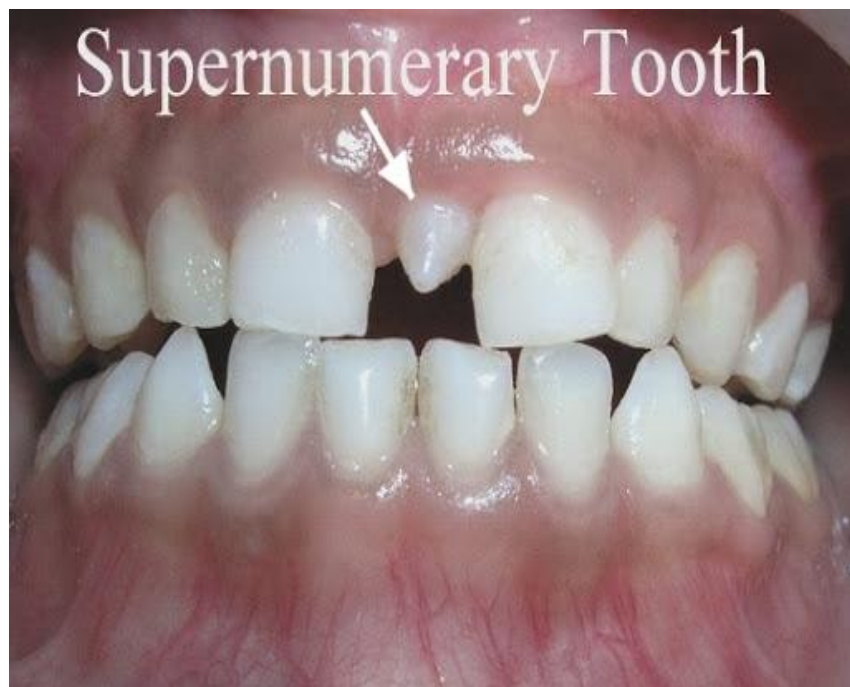


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- **8- Supernumerary tooth(Extra tooth)**
- These teeth maybe impacted or malposed and such teeth may predispose to malocclusion, periodontal disease, facial pain, esthetic problems, bony pathology and preventing eruption of other teeth.



- **9- Tooth in the line of fracture of the jaws**
- This tooth maybe extracted if:
 - a) It is a source of infection at the site of the fracture
 - b) The tooth itself is fractured.
 - c) Interfere with fracture reduction.
 - d) Interfere with healing of fracture

- **10- Teeth in relation with pathological conditions:-**

- They are indicated for extraction if they are involved in:

- a) Cyst formation.
- b) Neoplasm (tumor).
- c) Osteomyelitis (Infection of bone).
- d) Pyogenic granuloma

And the tooth interfere with complete surgical removal of the lesion the tooth should be extracted.

- **11- Root fragments (Retained root)**

- Retained roots may remain embedded in the bone without problems for a long period, but sometimes removal of such roots maybe necessary, for example,

1- root maybe at the sub mucosal level producing **recurrent ulceration** under the denture,

2- sometimes root fragments may be involved in initiation of **bony lesions** like osteomyelitis, cystic lesion or neoplasm,

3- if such fragments are in close relation to the **neurovascular bundle** (e.g. inferior dental nerve of the mandible)

4- the patient may complain of facial pain or numbness in the area supplied by that nerve.

As a general rule, very small fragments maybe left alone and that patient should be kept under periodic observation, and all other root fragments are indicated for removal.



- **12- Teeth prior to irradiation (before radiotherapy)**
- Teeth which cannot be kept in a sound condition should be removed before radiation because extraction of tooth in irradiated jaw with superadded infection will lead to develop osteoradionecrosis of the jaw bone which is serious complication and difficult to be treated.

- **13- Aesthetic considerations**

- Tooth with poor aesthetic due to severe **staining** (tetracycline, flourosis) or **hypoplasia** of enamel or dentine and they cannot be restored may be indicated for extraction.

14-Focal sepsis:-

Sometimes teeth or a tooth may appear sound clinically, but on radiographic examination the tooth may appear to be considered as a foci of infection (teeth associated with periapical pathology or periodontal problems), these teeth or tooth should be extracted in certain conditions e.g. heart surgery; heart valve replacement, kidney transplant, eye Surgery.

Contraindications

- In general, the contra-indications are divided into Two groups
- **1-Local contra-indications.**
- **2-Systemic contra-indications.**

1-Local contra-indications:

- 1- Acute and uncontrolled infections-
- Extraction in the presence of acute and uncontrolled infection may lead to spread of infection locally or systemically leading to many complications some of them are dangerous and life threatening (eg cavernous sinus thrombosis, mediastinitis, Ludwig's angina) and acute periapical abscess and facial abscess especially in medically compromised patient. In addition to that, limitation of mouth opening especially in lower wisdom tooth infection.

- 2- Previous radiotherapy
- Previous therapeutic radiation in oral and maxillofacial region for treatment of cancer lead to fibrosis and decreased vascularity of the tissue or area of extraction and end with a condition in the bone called osteoradionecrosis.



- 3- Teeth located within area of tumor
- Especially malignant tumor should not be extracted because extraction may lead to dissemination of the tumor, unhealed socket and intraoperative and postoperative a bleeding.

Systemic contra-indications

- Systemic contra-indications preclude extraction because the patient's systemic health is such that the ability to withstand the surgical work may be compromised. So extraction should be postponed until the severity of the problem has been resolved and maybe arranged after consultation with the physician to perform extraction safely without complications so caution is advised in the following conditions:-

Systemic contra-indications

- 1-Sever **uncontrolled metabolic disease**, eg, uncontrolled **diabetes**, end stage renal disease.
- 2- Uncontrolled **leukemias and lymphomas**.
- 3- Sever **uncontrolled cardiac disease**. eg, myocardial infarction, unstable angina pectoris, dysrhythmias.
- 4- Sever **uncontrolled hypertension**.

- 5-Pregnancy (some cases).
- 6- Uncontrolled bleeding disorder eg; hemophilia, platelet disorder, patient on anticoagulants.
- 7- Patients who take a variety of medications eg: patient on steroid and immunosuppressive drugs, cancer and chemotherapy.
- 8- Uncontrolled epilepsy.

